

Fundamental minority ICRF heating in ITER-EDA

The nominal $N=1$ reference case

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Abstract

The reference fundamental ITER-EDA case uses the same mass-3 minority at 50% concentration in deuterium as its second-harmonic counterpart, but heated at 30.45 MHz. It settles in 1.0 s of physical time at an effective temperature of 17.25 keV against a 12 keV bulk, remaining nearly isotropic at 55.7%. Despite absorbing *more* power than the $N=2$ case, 2.31 against 1.82 MW/m³, it heats the population as a whole rather than building a supra-thermal tail, and sheds 92% of that power to the background ions. It is a mild, ion-drag-dominated equilibrium.

1 Configuration

The plasma is identical to the $N=2$ reference: a tritium-like minority (mass 3, $Z=1$) in deuterium, both at 50% of $n_e = 1.5 \times 10^{20} \text{ m}^{-3}$, with $T_e = T_i = 12 \text{ keV}$ and $B_0 = 6 \text{ T}$. Only the wave differs (Table 1) — fundamental resonance at half the frequency, a smaller $k_\perp$, and field amplitudes some five times lower. The velocity domain is correspondingly smaller. Self-collisions use the non-linear Rosenbluth operator (`isc = -1`) with $\theta = 0.75$ (`icn = 1`).

Table 1: Case parameters. The run terminated on the `conv_diag` criteria at $t = 1.0 \text{ s}$ after 77 s of wall time, twelve times faster than the $N=2$ case.

harmonic	$N = 1$	grid	150×150 , quadratic in $v_\perp$
frequency	30.45 MHz	$v_\perp^{\max}$	$2.0 \times 10^7 \text{ m/s}$
$k_\perp, k_\parallel$	30, 2 m^{-1}	$v_\parallel$ range	$[-6, 6] \times 10^6 \text{ m/s}$
E_+	(250, 100) V/m	minority	mass 3, $Z = 1$, 50%
E_-	(750, 1) V/m	majority	deuterium, 50%

2 The steady distribution

The converged distribution (Fig. 1) is close to a Maxwellian that has simply been warmed. There is no pronounced perpendicular structure: the J_0 -weighted fundamental operator deposits its energy far more broadly in $v_\perp$ than the J_1 -weighted second harmonic, and the resulting distribution stays within a few per cent of isotropy throughout.

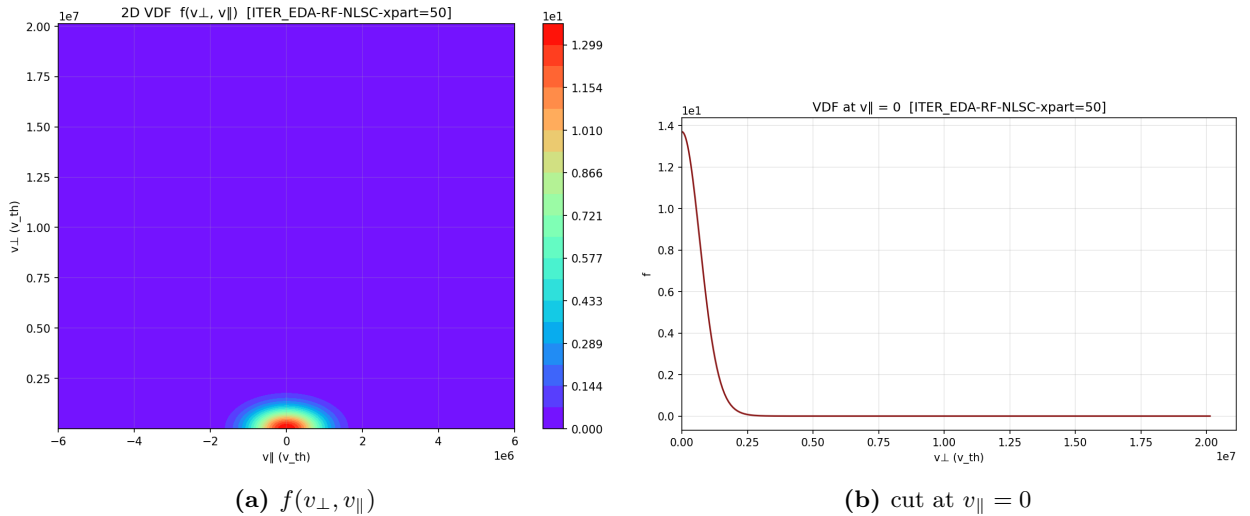
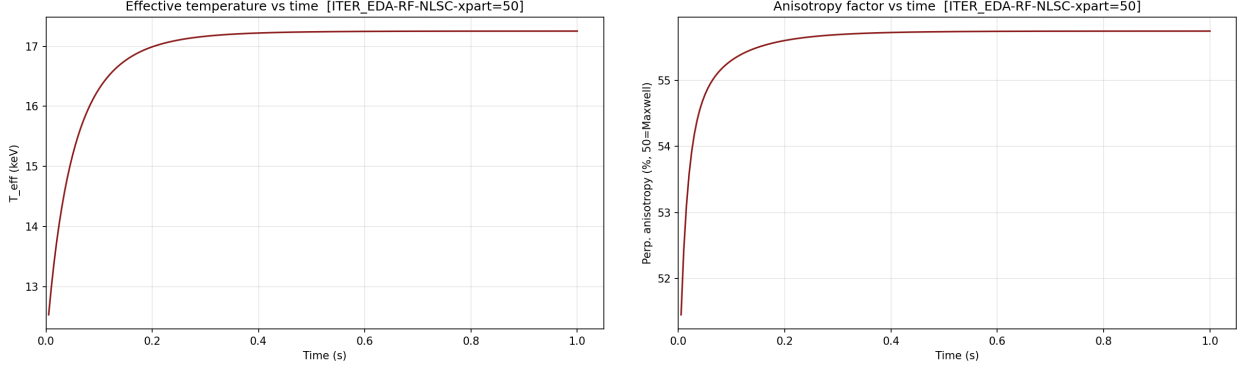


Figure 1: Converged minority distribution. Compare with the second-harmonic case, whose contours are strongly elongated in $v_\perp$; here they are almost circular.

3 Bulk heating rather than tail formation

The clearest signature of the difference between the two harmonics is not T_{eff} but its relation to the density-characteristic temperature T_n . Here T_{eff} rises from 12.53 to 17.25 keV and T_n follows it closely, from 12.46 to 16.23 keV (Fig. 2): the two agree to 6%, meaning the entire population has been heated. In the $N=2$ case the same pair reads 32.59 and 13.73 keV — a factor 2.4 apart — because there the energy sits in a tail that the second moment sees and the bulk does not.

The anisotropy tells the same story from another angle, rising only from 51.5% to 55.7% against the second harmonic's 73.9%. Both quantities saturate by $t \simeq 0.6$ s.



(a) effective temperature

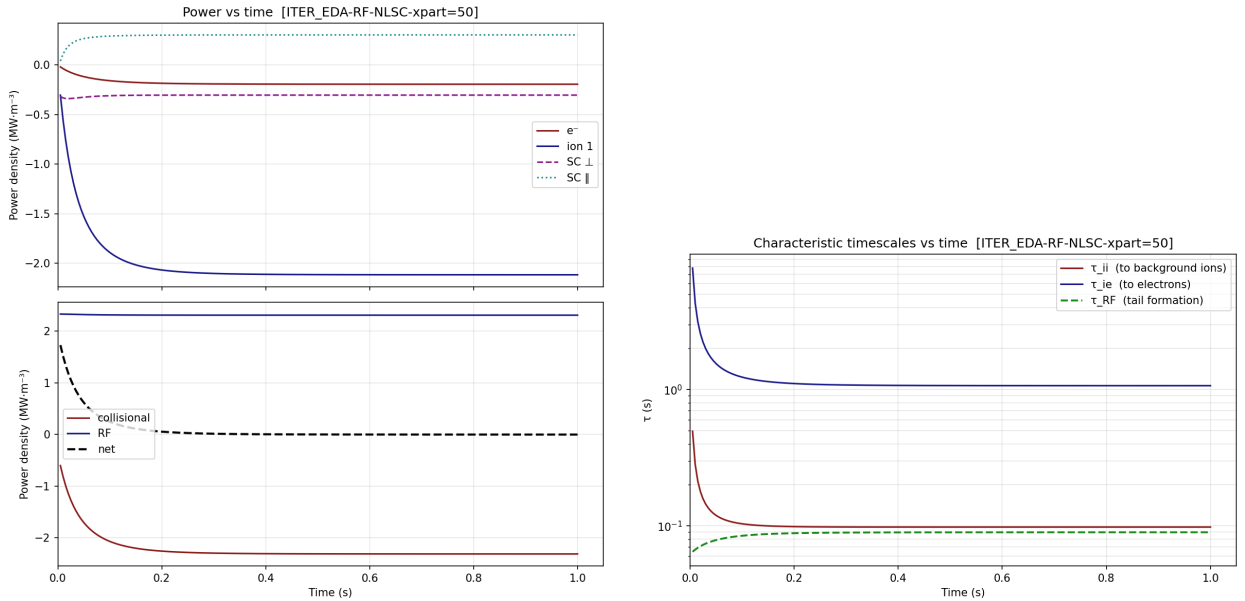
(b) perpendicular anisotropy

Figure 2: Approach to steady state; 50% anisotropy would be isotropic.

4 Power balance and timescales

The RF supplies 2.3118 MW/m³ and the collisions remove the same to within 0.001% (Fig. 3). The split is overwhelmingly to the ions: 2.118 MW/m³ against 0.194 to the electrons, or 92:8. With a critical energy of $E_c \simeq 106$ keV and a tail at 17 keV, the minority sits deep in the ion-drag regime and never approaches the ion-to-electron transition that the second harmonic reaches at 59:41.

Self-collisions again carry no net energy, moving 0.304 MW/m³ from the perpendicular direction to the parallel and closing to 0.004%. It is worth noting that this is *larger* than the 0.176 MW/m³ of the $N=2$ case, even though neglecting self-collisions costs only 5.6% here against 144% there: the magnitude of the self-collision term is no guide to how much the answer depends on it.



(a) power balance

(b) characteristic timescales

Figure 3: Power channels and the effective timescales $\tau = nT_{\text{eff}}/P$. The identity $1/\tau_{\text{RF}} = 1/\tau_{ii} + 1/\tau_{ie}$ holds to 0.001%.

As timescales, the RF fills on $\tau_{\text{RF}} = 0.090\text{ s}$ and the ions drain on $\tau_{ii} = 0.098\text{ s}$ — the two are within 8%, so the equilibrium is essentially a balance between the RF and ion drag alone. Electron drag, at $\tau_{ie} = 1.07\text{ s}$, is an order of magnitude slower and barely participates. That near-equality of τ_{RF} and τ_{ii} , against the factor 1.7 separating them at $N=2$, is why the fundamental sustains so much less supra-thermal population.

5 Summary and numerical quality

Table 2: Steady-state results, with the $N=2$ reference alongside.

	$N=1$	$N=2$
T_{eff}	17.25 keV	32.59 keV
T_n	16.23 keV	13.73 keV
anisotropy	55.75%	73.86%
P_{RF}	2.3118 MW/m ³	1.8166 MW/m ³
ion : electron split	92 : 8	59 : 41
τ_{RF}	0.0897 s	0.2156 s
τ_{ii}, τ_{ie}	0.098, 1.068 s	0.364, 0.528 s
convergence	1.0 s	12.5 s

Numerically this is the better-behaved of the two cases. It stops on the convergence criteria after 1.0 s, with a residual drift in T_{eff} of 0.008% over the final quarter of the run and a density change of +0.003% — forty times smaller than the $N=2$ case, since the self-collision operator’s slow particle gain has far less time to accumulate. Wall truncation of f is 9.6×10^{-27} of the peak in $v_{\perp}$ and 6.2×10^{-13} in $v_{\parallel}$, so the smaller velocity domain used here is amply sufficient.

Case **nominal#1**; figures from **figs/**, generated by **fp2d_plot.py** from the traces in **results/**.